

CSE113

Foundations of Computer Science

State University of New York, Korea

Previous lectures

Truth table

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Equivalence laws

Laws	Formula	Formula
Commutative laws	$p \wedge q \equiv q \wedge p$	$p \vee q \equiv q \vee p$
Associative laws	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$(p \vee q) \vee r \equiv p \vee (q \vee r)$
Distributive laws	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$

Propositional logic: A formal language to express facts and argue about them

Valid arguments

Premise ₁
Premise ₂
⋮
Premise _m
∴ Conclusion

Inference

Name	Rule	Name	Rule
Modus Ponens	$p \rightarrow q$ p ∴ q	Elimination	$p \vee q$ $p \vee q$ $\sim q$ $\sim p$ ∴ p ∴ q
Modus Tollens	$p \rightarrow q$ $\sim q$ ∴ $\sim p$	Transitivity	$p \rightarrow q$ $q \rightarrow r$ ∴ $p \rightarrow r$

Plan

- Predicate Logic, or propositions with Quantifiers
- Negation

Propositions with Quantifiers

Why quantifiers? To express “for all” and “there exists”

- Everyone can make mistake
- Nobody is perfect
- Every lock has a key
- There is a key for every lock
- Every nonzero real number has a reciprocal

CS example 1:

Software security

Question: Could the program print “bad”?

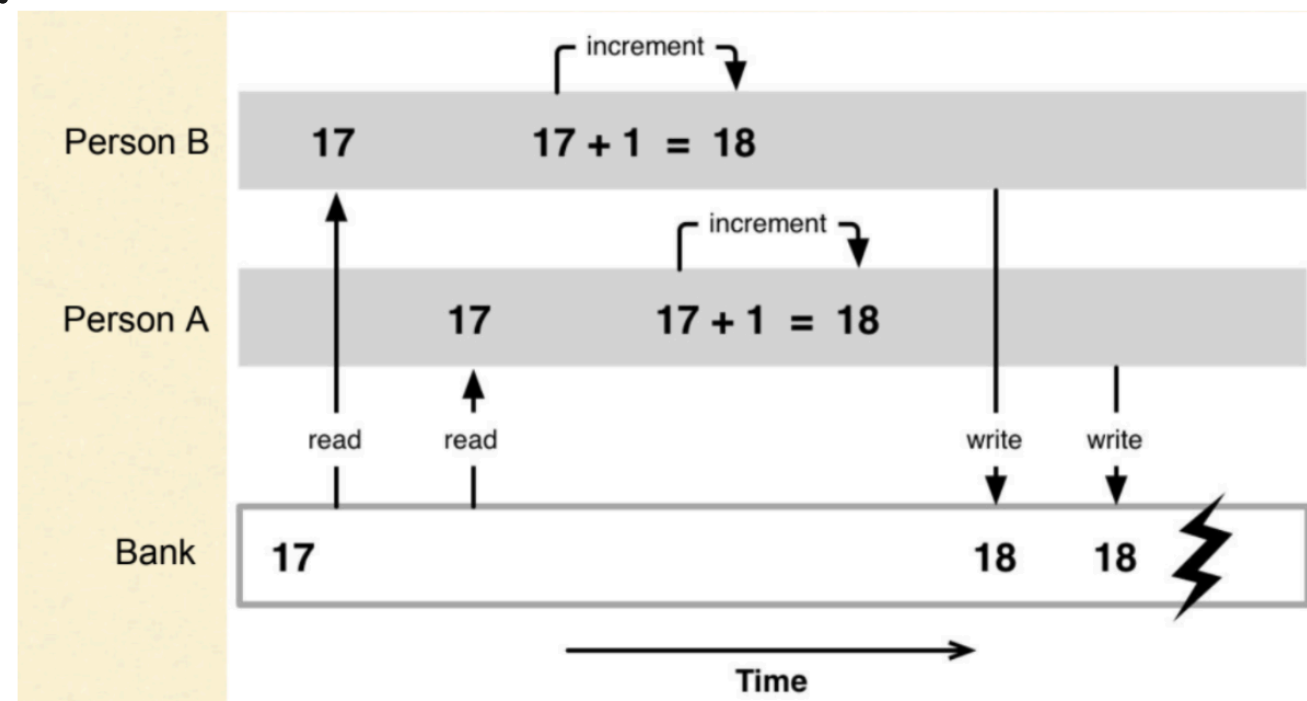
```
1  #include <stdio.h>
2
3  void f(int input) {
4      char a[8];
5      int b = 0;
6      a[input] = 1;
7
8      if (b == 0)
9          printf("good\n");
10     else
11         printf("bad\n");
12 }
```

Exisitential Statement

: there exists an integer n , such that $b \neq 0$ at line 8 when executing $f(n)$

CS example 2: Concurrency

Two persons are trying to deposit 1 dollar online into the same bank account.



Universal Statement:

For all CPU schedule s , A should not read while B intends to write.

Predicate

- A **propositional function** or **predicate** is a sentence that contains **one or more variables**
- A predicate is neither true nor false
- A predicate becomes a proposition when the variables are substituted with specific values
- The **domain** of a predicate variable is the set of all values that may be substituted for the variable

Examples

Symbol	Predicate	Domain	Propositions
$p(x)$	$x > 5$	$x \in \mathbb{R}$	$p(6), p(-3.6), p(0), \dots$
$p(x, y)$	$x + y$ is odd	$x \in \mathbb{Z}, y \in \mathbb{Z}$	$p(4, 5), p(-4, -4), \dots$

Universal statement

- Let $p(x)$ be a predicate and D be the domain of x
- A **universal statement** is a statement of the form

$$\forall x \in D, p(x)$$

- Forms:
 - “ $p(x)$ is true for all values of x ”
 - “For all x , $p(x)$ ”
 - “For each x , $p(x)$ ”
 - “For every x , $p(x)$ ”
 - “Given any x , $p(x)$ ”

Existential statement

- Let $p(x)$ be a predicate and D be the domain of x
- An **existential statement** is a statement of the form

$$\exists x \in D, p(x)$$

- Forms:
 - “There exists an x such that $p(x)$ ”
 - “For some x , $p(x)$ ”
 - “We can find an x , such that $p(x)$ ”
 - “There is some x such that $p(x)$ ”
 - “There is at least one x such that $p(x)$ ”

Examples

Universal st.s	Domain
$\exists x \in D, x^2 \geq x$	$D = \{1, 2, 3\}$
$\exists x \in \mathbb{R}, x^2 \geq x$	$\mathbb{R}$

Propositions with multiple quantifiers

$\forall x \in D, \exists y \in E$, such that $p(x, y)$

Every lock has a key.

Propositions with multiple quantifiers

$\exists x \in D, \forall y \in E$, such that $p(x, y)$

There is a key for every lock.

Propositions with multiple quantifiers

$\forall x \in D, \forall y \in E$, such that $p(x, y)$

“Give me a place to stand, and a lever long enough, and I will move the Earth.”

Propositions with multiple quantifiers

$\exists x \in D, \exists y \in E$, such that $p(x, y)$

“There is someone in a park sitting on a bench”.

Four Notes

Note 1:

the order of quantifiers matters

- Every lock has a key
- For any lock L , there exists a key K , such that K can unlock L .
- There is a key for every lock
- There exists a key K , such that for any lock L , K can unlock L .

Note 2: A commonly used notational equivalence

$$\forall x \in D, p(x)$$

Equivalent to

$$\forall x, x \in D \rightarrow p(x)$$

Example: All doctors wear glasses

- for all d , if d is a doctor, then d wears glasses
- Formally, if we define
 - D to be the set of doctors,
 - wear_glass to be a function that takes a person x as an input, and returns true if x wears glasses
- then the following two statements are considered the same

$$\forall d \in D, \text{wear_glass}(d).$$

$$\forall d, d \in D \rightarrow \text{wear_glass}(d)$$

Note 3: Universal conditional statement

- A **universal conditional statement** is of the form

$$\forall x, \text{ if } p(x) \text{ then } q(x)$$

Examples

- $\forall x \in \mathbb{R}, \text{ if } x > 2 \text{ then } x^2 > 4$
- $\forall$ real number x , if x is an integer then x is rational
 $\forall$ integer x , x is rational
- $\forall x$, if x is a square then x is a rectangle
 $\forall$ square x , x is a rectangle

Note 4: Implicit quantifiers

Examples

- If **a number** is an integer, then it is a rational number
Implicit meaning: $\forall$ number x , if x is an integer, x is rational
- **The number** 10 can be written as a sum of two prime numbers
Implicit meaning: $\exists$ prime numbers p and q such that $10 = p + q$
- If $x > 2$, then $x^2 > 4$
Implicit meaning: $\forall$ real x , if $x > 2$, then $x^2 > 4$

Definition

- Let $p(x)$ and $q(x)$ be predicates and D be the common domain of x . Then implicit quant. symbols $\Rightarrow, \Leftrightarrow$ are defined as:

$$p(x) \Rightarrow q(x) \equiv \forall x, p(x) \rightarrow q(x)$$

$$p(x) \Leftrightarrow q(x) \equiv \forall x, p(x) \leftrightarrow q(x)$$

Exercises

Exercise 1

translate to formal logic

Everyone can make mistake

Exercise 2

translate to formal logic

Nobody is perfect

Exercise 3

translate to formal logic

Every nonzero real number has a reciprocal

The reciprocal of 4 is $\frac{1}{4}$ (namely 0.25)

Exercise 4: Translate to formal logic

- Given:
 - S : set of students
 - $P(s)$: s passed the exam.
 - $W(s)$: s worked hard.
 - $C(s)$: s is in Professor Cho's class.
- **Every student in Professor Cho's class passed the exam**

Exercise 5: Translate to formal logic

- Given:
 - S : set of students
 - $P(s)$: s passed the exam.
 - $W(s)$: s worked hard.
 - $C(s)$: s is in Professor Cho's class.
- **Some students studied hard but did not pass the exam**

Exercise 6: Translate to formal logic

- Given:
 - S : set of students
 - $P(s)$: s passed the exam.
 - $W(s)$: s worked hard.
 - $C(s)$: s is in Professor Cho's class.
- **There are students who did not study hard but passed the exam**

Exercise 7: Translate to formal logic

- Given:
 - S : set of students
 - $P(s)$: s passed the exam.
 - $W(s)$: s worked hard.
 - $C(s)$: s is in Professor Cho's class.
- **All students who studied hard passed the exam.**

Exercise 8: Translate to formal logic

- Given:
 - S : set of students
 - $P(s)$: s passed the exam.
 - $W(s)$: s worked hard.
 - $C(s)$: s is in Professor Cho's class.
- **No student in Professor Cho's class failed the exam.**

Exercise 9: Translate to formal logic

- Given:
 - L: set of locks
 - K set of keys
 - unlock (k,l): k can unlock l
- **Some keys cannot unlock any lock.**

Negation

Negation of a universal statement

$$\sim (\forall x \in D, p(x)) \equiv \exists x \in D, \sim p(x)$$

Example: "All swans are white."

Negation: "There exists at least one swan that is not white."

Exercise

- Negate "Every phone on the table is turned off."

Negation of an existential statement

$$\sim (\exists x \in D, p(x)) \equiv \forall x \in D, \sim p(x)$$

Example: "There is a car in the parking lot that is electric."

Negation: "No car in the parking lot is electric", or "For every car c in the parking lot, c is not electric"

Exercise

- Negate "There is a person in the village who speaks Italian."

Summary

$$\sim (\forall x \in D, p(x)) \equiv \exists x \in D, \sim p(x)$$

$$\sim (\exists x \in D, p(x)) \equiv \forall x \in D, \sim p(x)$$

- Negation of a **universal** statement (“all are”) is logically equivalent to an **existential** statement (“there is at least one that is not”)

Negation of an **existential** statement (“some are”) is logically equivalent to a **universal** statement (“all are not”)

Exercise

- $\forall$ primes p , p is odd

Exercise

- $\exists$ triangle T , sum of angles of T equals 200°

Exercise

- No child is left behind

Common mistakes

Examples

- All mathematicians wear glasses

Negation (**incorrect**): No mathematician wears glasses

Negation (**correct**): There is at least one mathematician who does not wear glasses

- Some snowflakes are the same

Negation (**incorrect**): Some snowflakes are different

Negation (**correct**): All snowflakes are different

Negation of universal conditional statements

Definition

- Formally,

$$\begin{aligned}\sim (\forall x, p(x) \rightarrow q(x)) &\equiv \exists x, \sim (p(x) \rightarrow q(x)) \\ &\equiv \exists x, (p(x) \wedge \sim q(x))\end{aligned}$$

Negation of universal conditional statements

Definition

- Formally,

$$\begin{aligned}\sim (\forall x, p(x) \rightarrow q(x)) &\equiv \exists x, \sim (p(x) \rightarrow q(x)) \\ &\equiv \exists x, (p(x) \wedge \sim q(x))\end{aligned}$$

Examples

- $\forall$ real x , if $x > 10$, then $x^2 > 100$.
Negation: $\exists$ real x such that $x > 10$ and $x^2 \leq 100$.
- If a computer program has more than 100,000 lines, then it contains a bug.
Negation: There is at least one computer program that has more than 100,000 lines and does not contain a bug.

Negation of propositions with multiple quantifiers

$$\begin{aligned} & \sim (\forall x \text{ in } D, \exists y \text{ in } E \text{ such that } P(x, y)) \\ & \equiv \exists x \text{ in } D \text{ such that } \sim (\exists y \text{ in } E \text{ such that } P(x, y)) \\ & \equiv \exists x \text{ in } D \text{ such that } \forall y \text{ in } E, \sim P(x, y) \end{aligned}$$

$$\begin{aligned} & \sim (\exists x \text{ in } D \text{ such that } \forall y \text{ in } E, P(x, y)) \\ & \equiv \forall x \text{ in } D, \sim (\forall y \text{ in } E, P(x, y)) \\ & \equiv \forall x \text{ in } D, \exists y \text{ in } E \text{ such that } \sim P(x, y) \end{aligned}$$

Exercise

- Do some research: Formal definition of continuity of a real-valued function f on a point x
- Give a formal definition of f being discontinuous on x

Exercise: Translate to formal logic, and then negate

- Given:
 - $P(x)$: x is a person.
 - $Q(x)$: x is busy.
 - $R(x, y)$: x likes y .
- **All people are busy.**

Exercise: Translate to formal logic, and then negate

- Given:
 - $P(x)$: x is a person.
 - $Q(x)$: x is busy.
 - $R(x, y)$: x likes y .
- **Some people are not busy.**

Exercise: Translate to formal logic, and then negate

- Given:
 - $P(x)$: x is a person.
 - $Q(x)$: x is busy.
 - $R(x, y)$: x likes y .
- **Every person likes themselves.**

Exercise: Translate to formal logic, and then negate

- Given:
 - $P(x)$: x is a person.
 - $Q(x)$: x is busy.
 - $R(x, y)$: x likes y .
- **There's someone who doesn't like themselves.**

Exercise: Translate to formal logic, and then negate

- Given:
 - $P(x)$: x is a person.
 - $Q(x)$: x is busy.
 - $R(x, y)$: x likes y .
- **There's at least one person that everyone likes.**

Exercise: Translate to formal logic, and then negate

- Given:
 - $P(x)$: x is a person.
 - $Q(x)$: x is busy.
 - $R(x, y)$: x likes y .
- **Everyone likes at least one person.**

Exercise: Translate to formal logic, and then negate

- Given:
 - $P(x)$: x is a doctor.
 - $Q(x)$: x is busy.
 - $R(x, y)$: x likes y .
- **Some people don't like themselves.**